

UST v5 — New Theorems T29–T32

With Open Questions for CMB-S4 2028

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Confirmed Theorems

T#	Content	Predicted	Observed	$\Delta\%$
T29	Compact spectrum $\approx \sqrt{3}$	1.728845	1.732051	0.185
T30	Solar $\eta = 6\pi N_b C_{c_b}$	4.376223	4.383722	0.171
T32a	$\kappa \cdot dt_{\text{UST}} = \pi$ (exact)	π	π	0.000
T32b	$t_H \times N_b \approx \nu_{Cs-133}$	9.1920 Gyr	9.1926 GHz	0.007

Open Questions — Pending CMB-S4 2028

Three testable predictions from $RA = N_b - N_m = 9.38 \times 10^{-6}$:

$$A_s \approx 24 \times RA^2 = 24(N_b - N_m)^2 = 2.1116 \times 10^{-9} \quad \text{Planck 2018: } 2.099 \times 10^{-9} \quad \Delta\% = 0.60 \quad (1)$$

$$r = \frac{C_{c_b}}{24} = \frac{1 - N_b}{24} = 0.01527 \quad \text{BICEP/Keck: } r < 0.036 \quad \checkmark \quad (2)$$

$$f_{NL}^{\text{local}} = 0.000 \quad (\text{single scalar field, Planck 2018: } |f_{NL}| < 10) \quad (3)$$

Status: Factor of 24 lacks first-principles derivation. These remain open questions until CMB-S4 data (2028).

Falsification: If CMB-S4 measures $A_s \neq 2.111 \times 10^{-9}$ at 5σ , or $r > 0.020$, or $|f_{NL}| > 1$ — this mechanism is ruled out.